BPM2_ESOFT QUIZ 3_S2026_GD

# of Questions: 30

Question #: 1

A 51-year-old woman is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation, intermittent blood in the stool, and an unintentional weight loss 6.8-kg (15-lb). Physical examination shows pallor of the oral mucosa and conjunctivae. Colonoscopy shows a mass in the descending colon that bleeds easily. Biopsy of the mass shows cells with multiple mitoses and loss of architecture. Which of the following best characterizes the affected gastrointestinal structure in this patient?

A. Haustra coli
B. Well-developed brush border
C. Plicae circularis
D. Intestinal villi
E. Paneth cells

Question #: 2

A 56-year-old man comes to the physician because of a 3-month history of nausea, weight loss and abnormal bowel movements. He appears emaciated. Physical examination shows abdominal distension and a palpable mass in the right upper quadrant. CT scan of the abdomen shows a tumor in the right hepatic flexure of the colon. Based on the location of the tumor, which of the following groups of lymph nodes will most likely receive metastasizing cells from the cancerous lesion?

A. Internal iliac
B. Inferior mesenteric
C. Celiac
D. Superior mesenteric
E. Superficial inguinal

Question #: 3

A 75-year-old man comes to the physician for a follow-up examination. Two weeks ago, he underwent surgery for an abdominal aortic aneurysm. During surgery, the inferior mesenteric artery is excised. Which of the following arteries will most likely continue to supply blood to the distal colon through its anastomotic connections?

A. Middle colic
B. Ileal
C. Common hepatic
D. Jejunal
E. Splenic

Question #: 4

A 22-year-old African American man comes to the emergency department because of sudden-onset back pain, fatigue and dark urine. Three days ago, he was prescribed trimethoprim-sulfamethoxazole for a urinary tract infection. Physical examination shows scleral icterus and pale mucus membranes. Results of laboratory studies are shown in the table. The activity assay of a specific enzyme shows decreased enzymatic activity. The coenzyme formed from the deficient enzyme is most likely required for which of the following metabolic processes?

A. Pyruvate to oxaloacetate
B. Alpha-ketoglutarate to succinyl CoA
C. HMG CoA to mevalonate
D. Fatty acyl CoA to acetyl CoA
E. HMG CoA to acetoacetate

Hemoglobin	8.5 g/dL (normal: 13.5 – 17.5)
Reticulocytes	7% (normal: 0.5% - 1.5%)
Urinalysis **Blood** **Red blood cells**	 Moderate 0/hpf
Liver function studies **Total bilirubin** **Direct bilirubin**	 3.5 mg/dL (0.1 – 1.0 mg/dL) 0.8 mg/dL (0.0 – 0.3 mg/dL)
Lactate dehydrogenase	700 U/L (45 – 200 U/L)
Peripheral smear	Bite cells and red blood cell inclusions on crystal violet staining

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Question #: 5

A 45-year-old man comes to the physician for a health maintenance examination. He says he had a burger with fries and a soft drink for lunch 10 minutes ago and is extremely drowsy. Physical examination shows no abnormalities. The physician explains that his metabolism is increased and there is increased activity of specific biochemical pathways. Which of the following enzymes and its corresponding substrate is most likely utilized during his current state?

A. I
B. II
C. III
D. IV
E. V
F. VI
G. VII
H. VIII

	Enzyme activity	Substrate
I	Pyruvate carboxylase	Pyruvate
II	Hormone sensitive lipase	Adipose TAGs
III	Glycogen phosphorylase	Glycogen
IV	Fatty acid synthase complex	Palmitate
V	Succinyl CoA: acetoacetate CoA transferase (Thiophorase)	Succinyl CoA
VI	Citrate Lyase	Citrate
VII	Glycogen synthase	Lactate
VIII	Mitochondrial HMG CoA Synthase	Acetyl CoA

Question #: 6

A 17-year-old girl is brought to the physician by her parents because of a 1-day history of right flank pain and red-colored urine. Imaging studies show an opaque mass in her right kidney. Urinalysis shows the presence of RBCs and hexagonal crystals. Deficiency of which of the following transporters is most likely responsible for this patient's condition?

A. Basic amino acid transporter
B. Branched chain amino acid transporter
C. Aromatic amino acid transporter
D. Acidic amino acid transporter
E. Neutral amino acid transporter

Question #: 7

A 10-day-old boy is brought to the emergency department by his parents because of a 6-hour history of lethargy, poor feeding and progressive unresponsiveness. He was born by vaginal delivery at-term and there were no abnormalities at birth. Laboratory studies show markedly elevated blood ammonia levels and respiratory alkalosis. Genetic tests show an inborn error of metabolism. Which of the following enzyme deficiency disorders is most likely in this patient?

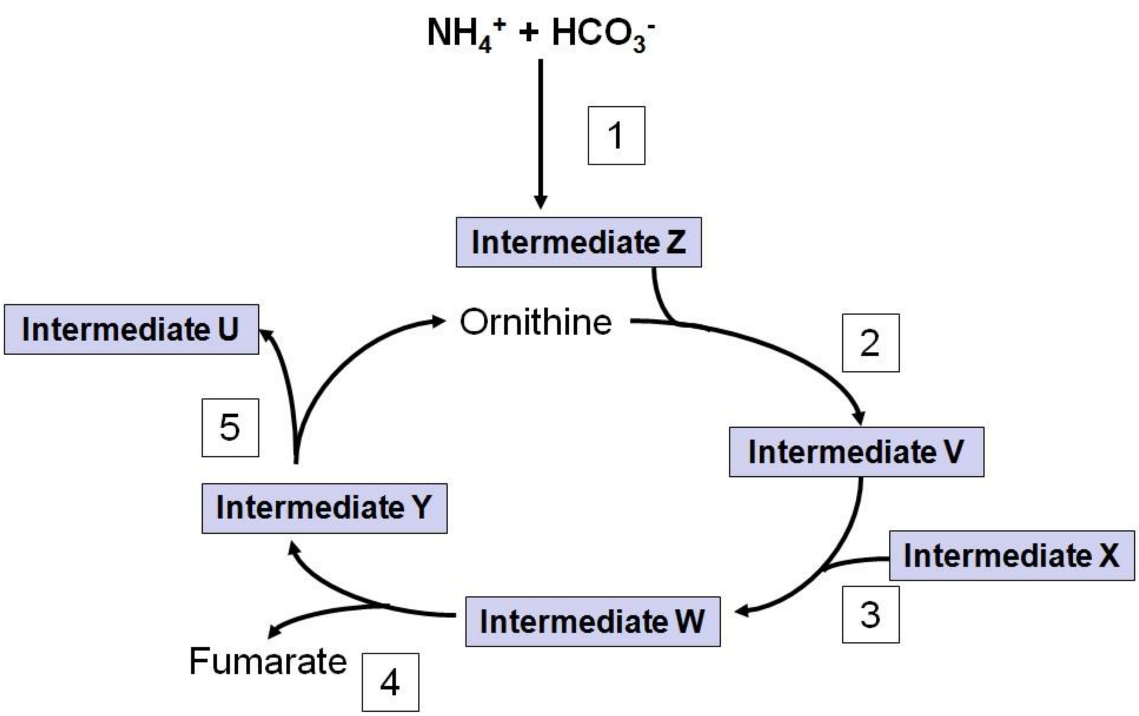
A. Glucose 6-phosphatase
B. Medium chain acylCoA dehydrogenase
C. Carnitine palmitoyl transferase-I (CPT-I)

D. Carbamyl phosphate synthetase-I (CPS-I)
E. Bilirubin UDP-glucuronyl tranferase

Question #: 8

A 4-month-old boy is brought to the emergency department by his parents because of a 1-day history of seizures. An inborn error of metabolism involving the pathway that is deficient in this patient is shown. Laboratory studies show high blood levels of argininosuccinate. Which of the following best represents the intermediate accumulating in this patient?

A. Intermediate U
B. Intermediate Z
C. Intermediate U
D. Intermediate W
E. Intermediate V



Question #: 9

A 3-month-old boy is brought to the physician because of a 5-day history of persistent crying and severe skin blisters. Physical examination shows blisters all over his body in various stages of healing. Urinalysis shows red color urine and elevated levels of porphyrins type I. Which of the following is the most likely diagnosis?

A. Lead poisoning
B. Pyridoxine deficiency
C. Acute intermittent porphyria
D. Congenital erythropoietic porphyria
E. Porphyria cutanea tarda
F. Gilbert syndrome

Question #: 10

A 4-week-old boy is brought to the physician by his mother because of a 3-day history of vomiting and failure to thrive. He is exclusively breast-fed. The mother says that vomiting usually occurs within an hour of breast feeding. Physical examination shows jaundice and an enlarged liver with a palpable lower border of the liver. He has a developmental delay. The blood glucose level is 50 mg/dL. Urinalysis shows the presence of a reducing sugar, that is not glucose. Which of the following enzyme deficiency disorders is the most likely diagnosis in this infant?

A. β-galactosidase
B. Fructokinase
C. Galactokinase
D. Galactose 1-phosphate uridyltransferase
E. Aldolase B
F. Branched chain keto acid dehydrogenase

Question #: 11

A 25-year-old woman is brought to the physician by her friend because of a 20-day history of starving herself. Physical examination shows muscle atrophy and poor grip strength. She is observed to have muscle protein catabolism. Which of the following amino acids most likely forms only acetyl CoA or ketone bodies on catabolism?

A. Alanine
B. Tyrosine
C. Glutamine
D. Aspartate
E. Leucine
F. Glycine

Question #: 12

A 2-month-old girl is brought to the physician by her parents because of a 2-day history of intermittent seizures. The mother says the urine has a mousy (musty) odor. Physical examination shows developmental delay. She is prescribed a special diet lacking an amino acid. Her mother is also advised to avoid aspartame in the infant's diet. Which of the following inherited disorders of metabolism is the most likely diagnosis?

A. Alkaptonuria
B. Phenylketonuria
C. Homocystinuria
D. Hyperammonemia type I
E. Maple syrup urine disease
F. Medium chain acyl CoA dehydrogenase deficiency
G. Galactosemia
H. Lactose intolerance

Question #: 13

A 25-year-old man has been starving for the past 20 days and shows significant denovo synthesis from amino acids. Which of the following amino acids would directly form oxaloacetate as a precursor of glucose by gluconeogenesis?

A. Alanine
B. Tyrosine
C. Glutamine

D. Aspartate
E. Leucine
F. Glutamate

Question #: 14

Metabolic changes in a 22-year-old fasting woman is being studied. She only sips approximately 100 ml of apple juice throughout day 1 of fasting. Which of the following is most likely observed on the second day of the fast?

A. The rate of β-oxidation is decreased in the liver
B. The activity of lipoprotein lipase is increased
C. Hepatic malonyl CoA levels are increased
D. CPT-1 activity is increased in the liver
E. Glycerol is utilized for energy generation in the adipose tissue
F. Hormone sensitive lipase is in the dephosphorylated form
G. Mitochondrial HMG-CoA synthase activity is low
H. Hepatic fructose 2,6 bisphosphate levels are elevated

Question #: 15

A 28-year-old woman comes to the physician because of a 3-month history of tiredness and easy fatigability. Physical examination shows conjunctival pallor. Laboratory studies show a hemoglobin of 9.2 g/dL (reference range 11.4-16.0 g/dL). A peripheral blood film shows macrocytic, megaloblastic anemia. Further tests show that almost all the folate is in the methyl-tetrahydrofolate form. Which of the following additional laboratory findings is most likely in this patient?

A. Increased serum protoporphyrin levels
B. Increased serum methylmalonate levels
C. Increased urinary copper excretion
D. Low serum ferritin levels
E. Decreased red cell transketolase activity

Question #: 16

A 43-year-old man comes to the physician because of a 6-month history of brownish-black discoloration of his ears and nose. He says it has slowly progressed in the last year and has become quite apparent in the past few weeks. He has a history of dark urine since birth. Accumulation of which of the following compounds is most likely responsible for this clinical finding?

A. Melanin
B. Homogentisic acid
C. Homocysteine
D. Coproporphyrin-III
E. Porphobilinogen
F. Cystine

Question #: 17

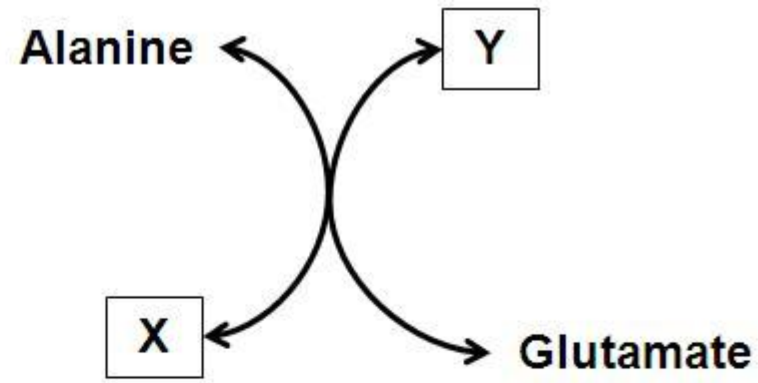
A 10-day-old boy is brought to the emergency department by his parents because of a 6-hour history of lethargy, poor feeding and progressive unresponsiveness. He was born by vaginal delivery at-term and there had no abnormalities at birth. Laboratory studies show markedly elevated blood ammonia levels and respiratory alkalosis. Genetic tests show an inborn error of metabolism. He is prescribed a drug to facilitate removal of excess nitrogen from his body. Which of the following drugs was most likely administered to this patient?

A. N-acetyl cysteine
B. Acetoacetic acid
C. Glutamine
D. Benzoic acid
E. Aspartate
F. Carnitine
G. Hemin
H. L-DOPA

Question #: 18

An investigator studying amino acid metabolism isolates an enzyme from cultured hepatocytes. The reaction catalyzed by the enzyme is shown. Which of the following best describes this reaction?

A. X represents ammonia and Y represents glutamine
B. X and Y are ketoacids of alanine and glutamate, respectively
C. The coenzyme required for this reaction is NAD^+
D. Patients with thiamine deficiency have low activity of this enzyme
E. X is alpha-ketoglutarate and Y is oxaloacetate



Question #: 19

An investigator is studying the regulation of an enzyme-catalyzed reaction shown. The activity of the enzyme is observed under different metabolic conditions. Which of the following most likely inhibits the activity of this enzyme?
HMG CoA → Mevalonate

A. Phosphorylation
B. Acetyl-CoA
C. Cholesterol ester
D. Citrate
E. Dephosphorylation
F. NADPH
G. Insulin

Question #: 20

A 23-year-old man comes to the physician because of a 2-day history of worsening bloating, diarrhea, dehydration and abdominal pain. He says the symptoms occur a few hours after eating meals with cereal, milk and sugar. Four days earlier, he had an episode of seafood poisoning with severe diarrhea, dehydration and abdominal pain. Which of the following is the best explanation for the worsening of the symptoms in this patient?

A. Increased activity of enteropeptidase
B. Decreased secretion of pancreatic amylase
C. Decreased activity of intestinal disaccharidases
D. Decreased secretion of bile salts by the gall bladder
E. Increased secretion of gastrin by the stomach

Question #: 21

A 50-year-old woman comes to the physician because of a 3-month history of weight loss. Physical examination shows conjunctival icterus and tenderness in the right epigastric region. Laboratory studies show elevated levels of serum total and direct bilirubin. Indirect bilirubin levels are normal. Serum ALP levels are increased; but serum amylase and lipase levels are normal. Ultrasound examination of the abdomen shows gall stones obstructing the common bile duct. Which of the following steps is most likely directly affected as a consequence?

A. Activation of pepsinogen to pepsin
B. Digestion of dietary starch
C. Absorption of dietary triacylglycerol
D. Formation of chylomicrons
E. Activation of trypsinogen to trypsin
F. Absorption of dietary vitamin B12
G. Digestion of disaccharides

Question #: 22

A 16-year-old boy is brought to the physician by his parents because of a 2-day history of yellowing of the eyes and skin. Two days earlier, he was prescribed a sulfa drug for a respiratory infection. A provisional diagnosis of inherited glucose 6-phosphate dehydrogenase deficiency is made. Which of the following sets of laboratory results will most likely be found in this patient?

A. I
B. II
C. III
D. IV
E. V

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Unconjugated	Absent	Decreased
II	Unconjugated	Present	Increased
III	Unconjugated	Absent	Increased
IV	Conjugated	Absent	Normal
V	Conjugated	Present	Decreased

Question #: 23

A 25-year-old man comes to the physician because of a 1-month history of intermittent abdominal cramps and bloating. History shows that the symptoms typically appear following meals rich in carbohydrates. Physical exam shows no abnormalities. Genetic testing shows a mutation in a gene which encodes for an enzyme that breaks down disaccharides to monosaccharides for absorption. Which of the following is most likely the typical localization of this enzyme?

A. Digestive organelles of hepatocytes
B. Cytoplasm of enteroendocrine cells
C. Microvilli of enterocytes
D. Glands of the intestinal submucosa
E. Lumen of the stomach

Question #: 24

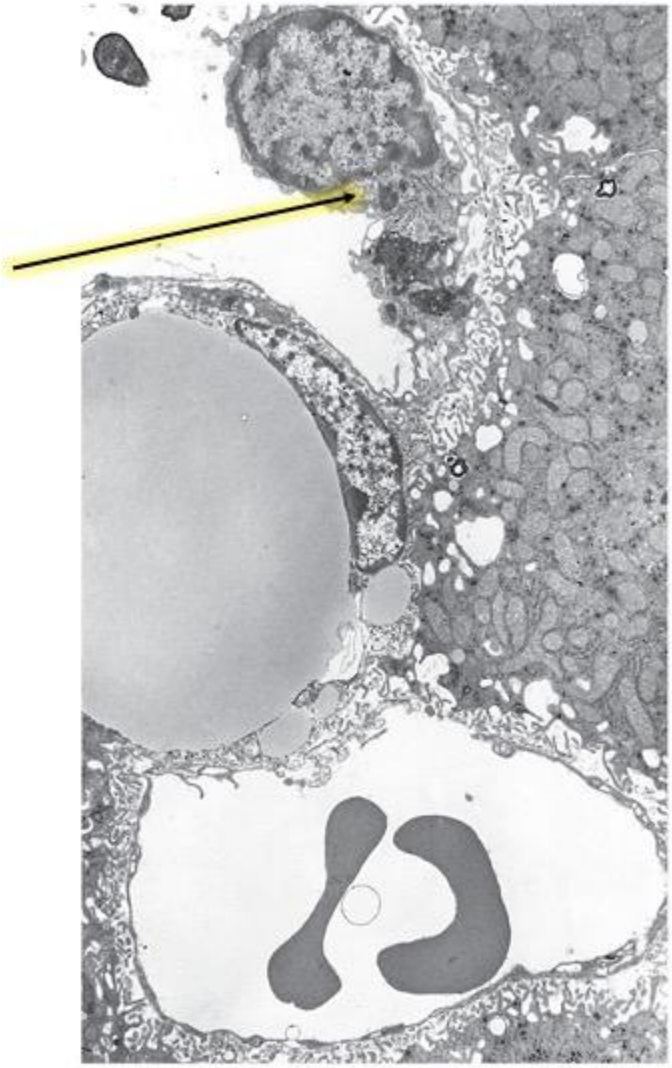
A 56-year-old woman is brought to the emergency department because of a 6-week history of fatigue, dizziness, and intermittent chest pain. History shows that she has Graves' disease. Her blood pressure is 90/60 mmHg and pulse is 120/min. Physical examination shows pale conjunctivae and oral mucosa and macroglossia (enlarged tongue). Laboratory studies of her blood show a hemoglobin level of 11 g/dL (normal range: 12-16 g/dL) and a vitamin B12 level of 165ng/mL (normal range 200-900 ng/mL). Which of the following is most likely to be an additional finding and a potential cause of the abnormalities seen in this patient?

A. Decreased lysosomal functions
B. Increased secretion of chymotrypsin
C. Antibodies against intrinsic factor
D. Increased secretion of mucus
E. Hyperplasia of chief cells

Question #: 25

A 27-year-old woman comes to the physician because of a 1-week history of loss of appetite, nausea, weakness, and dark urine. Her temperature is 38ºC (101ºF), pulse is 100/min and respirations are 22/min. Physical examination shows right upper quadrant abdominal tenderness. Laboratory studies of the blood show antibodies against hepatitis B virus. A diagnosis of acute hepatitis B infection is made. During this disease there is initial activation of the cells shown on the attached micrograph. Which of the following is the most likely function of these cells?

A. Production of bilirubin
B. Phagocytosis of red blood cells
C. Storage of vitamin A as retinyl esters
D. Degradation of exogenous toxins and drugs
E. Synthesis of steroid hormones



Question #: 26

A 64-year-old man comes to the physician because of a 3-week history of epigastric pain, abdominal discomfort, and weight loss. Physical examination shows no abnormalities. Laboratory studies of gastric juice show an abnormally high gastric acid output. The serum gastrin level is not elevated. Gastric mucosal studies show increased production of histamine in the stomach. Which of the following cells is the source of this substance?

A. ECL cells
B. S cells
C. Chief cells
D. D cells

E. G cells

Question #: 27

A 74-year-old woman comes to the physician for a follow-up examination. Six months ago, she was diagnosed with an early stage of gastric carcinoma in the upper stomach and a partial gastrectomy involving the fundal region of the stomach was performed. Physical examination shows no abnormalities. Which of the following functions of the stomach is most likely affected in this patient?

A. Acid secretion
B. Receptive relaxation
C. Basic electrical rhythm
D. Migrating motor complex
E. Mixing and grinding

Question #: 28

A 19-year-old man participates in a study related to the motility of the alimentary tract. He is given a single dose of the hormone motilin, which triggers peristaltic waves that start from the duodenum and move towards the caecum. The options of contraction and relaxation of smooth muscles in oral and caudal direction to the food are shown in the attached table. Which of the following options of smooth muscle contraction and relaxation is most likely during intestinal peristalsis?

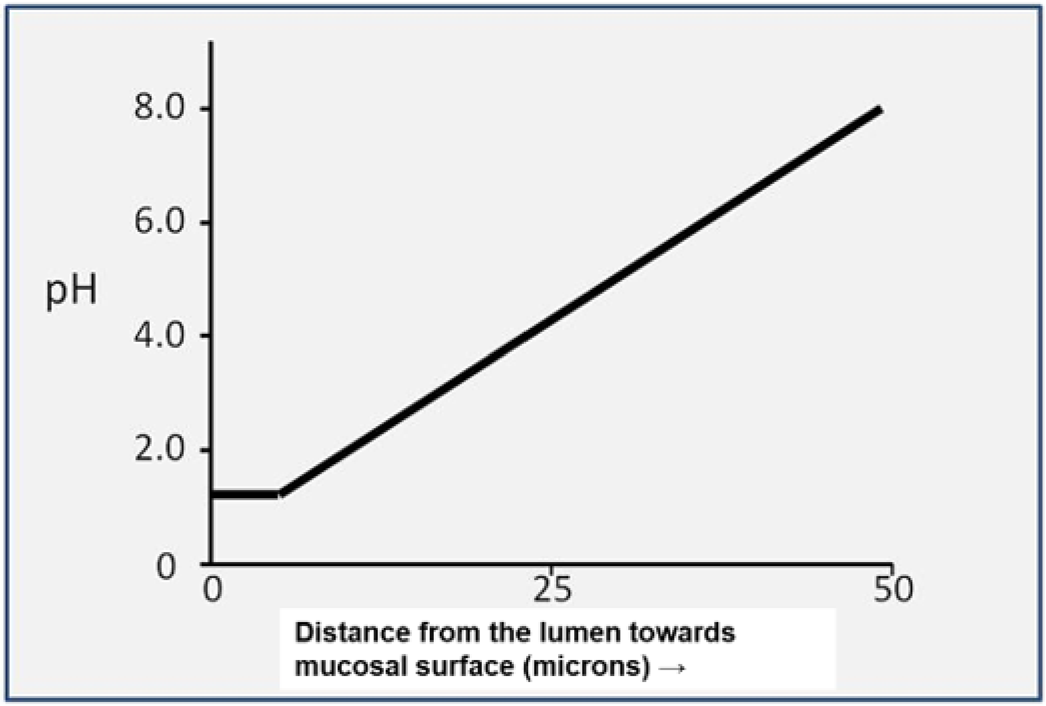
A. A
B. B
C. C
D. D

	Smooth Muscle Oral of the Food	Smooth Muscle Caudal of the Food
A	Contracted	Contracted
B	Relaxed	Contracted
C	Contracted	Relaxed
D	Relaxed	Relaxed

Question #: 29

A 22-year-old man volunteers to be a subject in a study on luminal fluid pH in the stomach. A pH-sensitive microelectrode is slowly advanced from the lumen of the stomach deep into the gastric mucosal surface. Results of change in pH as the microelectrode moves from the lumen deep inside is shown in the attached figure. It has been observed that the pH increases as the electrode approaches the gastric surface. Which of the following secretory changes best explains the observed increase in pH of the fluid?

A. Increased release of acetylcholine from the vagus nerve
B. Release of gastrin from the G cells
C. Secretion of bicarbonate ions by the mucus cells in the surface epithelium
D. Secretion of H^+ by mucus cells in the surface epithelium
E. Release of secretin from duodenal S-cells



Question #: 30

A 30-year-old woman comes to the physician because of a 6-month history of progressive difficulty swallowing and a 4-kg (8.8-lb) weight loss. Physical examination shows no abnormalities. A manometric study is conducted to examine pressure generation along the length of her esophagus. Manometry shows that esophageal contractions in response to a swallow are poorly synchronized and the pressure in the lower esophageal sphincter remains elevated at rest as well as during swallowing. Which of the following is the most likely diagnosis?

A. Gastroesophageal reflux disease
B. Achalasia
C. Hiatal Hernia
D. Scleroderma
E. Barrett's esophagus